Limits. Part 2

In this session[9] we are going to learn how to compute limits with a numerical approach and understand the main difference of computing the limit and just evaluate a function.

By evaluating a function, it answers to the question "*What is the function's value at this point?*" and it just inserting a number directly to the algebraic expression. On the other hand, taking a Limit allows to answer to "*What value does the function approach as we get arbitrarily close to a point?*" And the function doesn't even have to be defined at the point itself.

For example, let's consider

$$f(x) = \frac{x^2-1}{x-1}.$$

We already know that $x = 1$ is going to be undefined, becuase we are going to reach $0/0$. However, with limits, we can known the value that the function tends to.

In order to known the value that the function is approaching when the domain goes to 1, let's compute the limit using a numerical method. First lets build a table with values that are near 1, for instance,

x	$f(x) = \frac{x^2-1}{x-1}$
1.1	$\frac{(1.1)^2-1}{1.1-1} = \frac{1.21-1}{0.1} = \frac{0.21}{0.1} = 2.1$
1.01	$\frac{(1.01)^2-1}{1.01-1} = \frac{1.0201-1}{0.01} = \frac{0.0201}{0.01} = 2.01$
1.001	$\frac{(1.001)^2-1}{1.001-1} = \frac{1.002001-1}{0.001} = \frac{0.002001}{0.001} = 2.001$
1.000001	$\frac{(1.000001)^2-1}{1.000001-1} = \frac{1.000002000001-1}{0.000001} = \frac{0.000002000001}{0.000001} = 2.000001$

This is computing the limit as $x \to 1$ from the right to the left. We can also compute the limit as $x \to 1$ for the left to the right, as follows,

x	$f(x) = \frac{x^2-1}{x-1}$
0.9	$\frac{(0.9)^2-1}{0.9-1} = \frac{0.81-1}{-0.1} = \frac{-0.19}{-0.1} = 1.9$
0.99	$\frac{(0.99)^2-1}{0.99-1} = \frac{0.9801-1}{-0.01} = \frac{-0.0199}{-0.01} = 1.99$
0.999	$\frac{(0.999)^2-1}{0.999-1} = \frac{0.998001-1}{-0.001} = \frac{-0.001999}{-0.001} = 1.999$
0.999999	$\frac{(0.999999)^2-1}{0.999999-1} = \frac{0.999998000001-1}{-0.000001} = \frac{-0.000001999999}{-0.000001} = 1.999999$

With this we can say that the "*limit*" tells us the value the function is trying to be at $x = 1$, even though it never actually reaches it there.

[9] January 22

We can use the numerical approch to compute the limit of the example of the previous session,

$$\lim_{h\to 0}[29.4+4.9h]$$

h	$29.4+4.9h$
0.1	29.89
0.01	29.449
0.000001	29.4000049

As $h \to 0$,
$29.4+4.9h \to 29.4$
$\lim_{h\to 0}(29.4+4.9h) = 29.4$

Class Activity

Compute the limit with the numerical approach of the following functions,

$$f(x) = \frac{1}{x} \quad x \to 0$$
$$g(x) = \frac{x^2-4}{x-2} \quad x \to 2$$
$$h(x) = \frac{\sin(x)}{x} \quad x \to 0$$